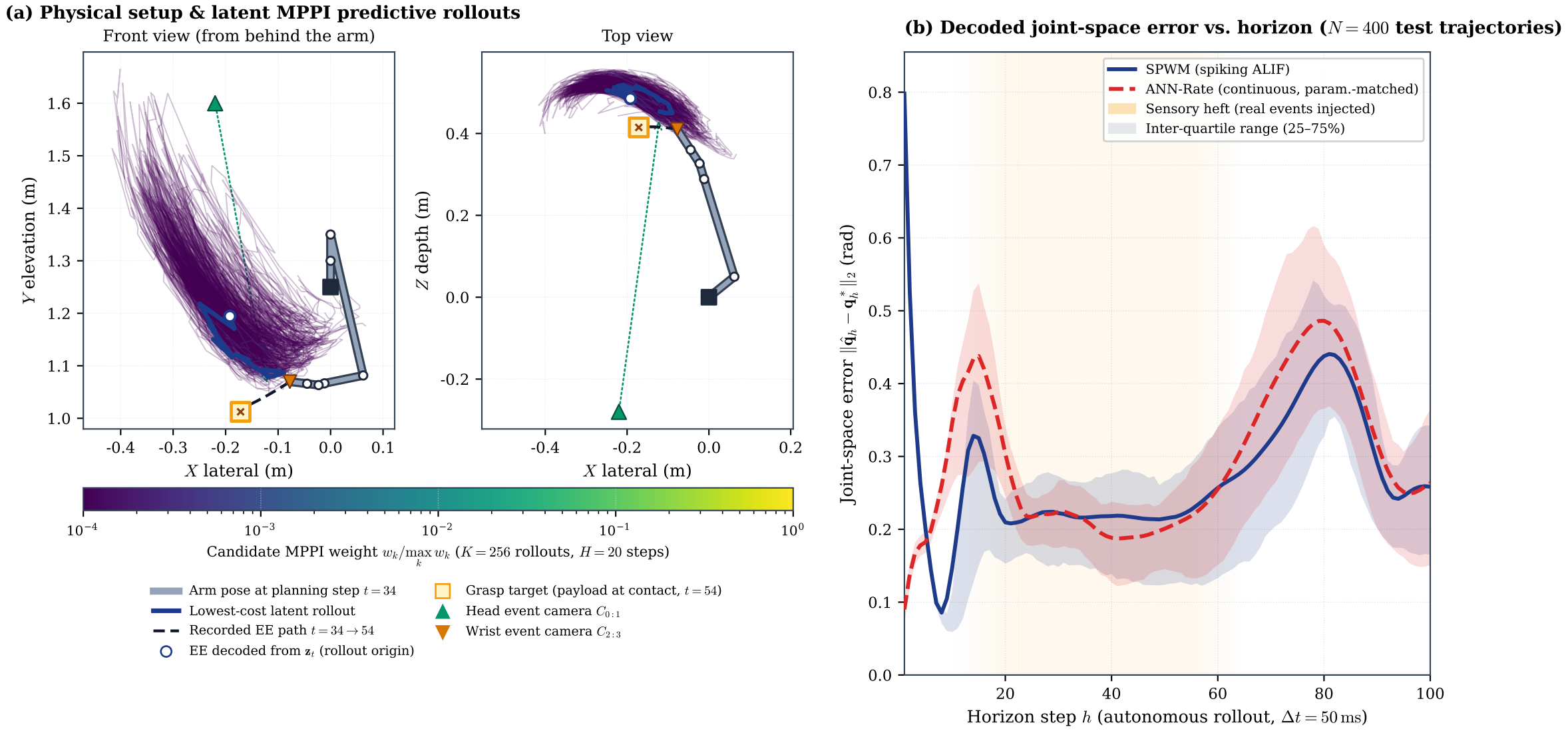

(a) Physical setup & latent MPPI predictive rollouts
Front view (from behind the arm)
Top view
Y elevation (m)
Z depth (m)
X lateral (m)
X lateral (m)
Candidate MPPI weight $w_k/\max_k w_k$ ($K = 256$ rollouts, $H = 20$ steps)
Arm pose at planning step $t = 34$
Lowest-cost latent rollout
Recorded EE path $t = 34 \to 54$
EE decoded from $\mathbf{z}_t$ (rollout origin)
Grasp target (payload at contact, $t = 54$)
Head event camera $C_{0:1}$
Wrist event camera $C_{2:3}$
(b) Decoded joint-space error vs. horizon ($N = 400$ test trajectories)
SPWM (spiking ALIF)
ANN-Rate (continuous, param.-matched)
Sensory heft (real events injected)
Inter-quartile range (25–75%)
Joint-space error $\|\hat{\mathbf{q}}_h - \mathbf{q}_h^*\|_2$ (rad)
Horizon step h (autonomous rollout, $\Delta t = 50$ ms)